PREDICTING NEW RELATIONSHIPS BETWEEN GENES AND HPO TERMS

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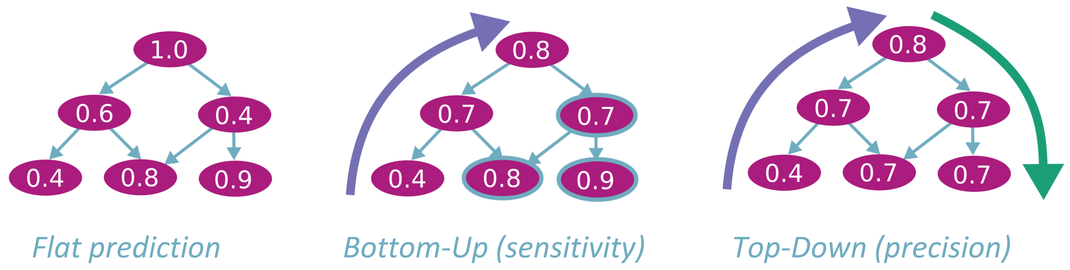
INTRODUCTION

The prediction of human gene–phenotype associations is a fundamental step towards the discovery of novel genes associated with human disorders. In this context the Human Phenotype Ontology (HPO) provides a standard categorization of phenotypes associated with human disease [1]. Most of the methods proposed in literature for new associations between genes and HPO terms are 'hierarchy-unaware' of the ontology, which causes inaccurate and biologically contradictory predictions.

In 2017 we published highly-modular hierarchical ensemble approaches that can be used to enhance the prediction of any flat learning method by taking the structure of the ontology into account [2]. Here we present the algorithm, DEScendant Classifier ENSemble (DESCENS), which considers a constant contribution of all the positive descendants of each node. In addition, we introduce a well documented software library, HEMDAG, for new gene–term predictions given an ontology and a gene network.

ALGORITHM

After 'flat' learning (e.g. using a SVM), the algorithm implements a step to combine the predictions hierarchically. First, a bottom-up approach increases the sensitivity and second, a top-down approach increases the precision.



In the bottom-up step, the ensemble decision is modified by averaging the local prediction of a node *i* with that of its positive children ϕ_i:

$$\bar{y}_i := \frac{1}{1+|\phi_i|}\left(\hat{y}_i + \sum_{j\in\phi_i} \bar{y}_j\right), \phi_i = \{j \in child(i) | \bar{y}_j > \hat{y}_i\}$$

DESCENS: to enhance the contribution of the most specific nodes we can consider all positive descendants:

$$\bar{y}_i := \frac{1}{1+|\Delta_i|}\left(\hat{y}_i + \sum_{j\in\Delta_i} \bar{y}_j\right), \Delta_i = \{j \in desc(i) | \bar{y}_j > \hat{y}_i\}$$

DESCENS-τ: $\tau \in [0,1]$ to balance the contribution between ϕ_i and δ_i, $\delta_i = \Delta_i \setminus \phi_i$:

$$\bar{y}_i := \frac{\tau}{1+|\phi_i|}\left(\hat{y}_i + \sum_{j\in\phi_i} \bar{y}_j\right) + \frac{1-\tau}{1+|\delta_i|}\left(\hat{y}_j + \sum_{j\in\delta_i} \bar{y}_j\right)$$

REFERENCES

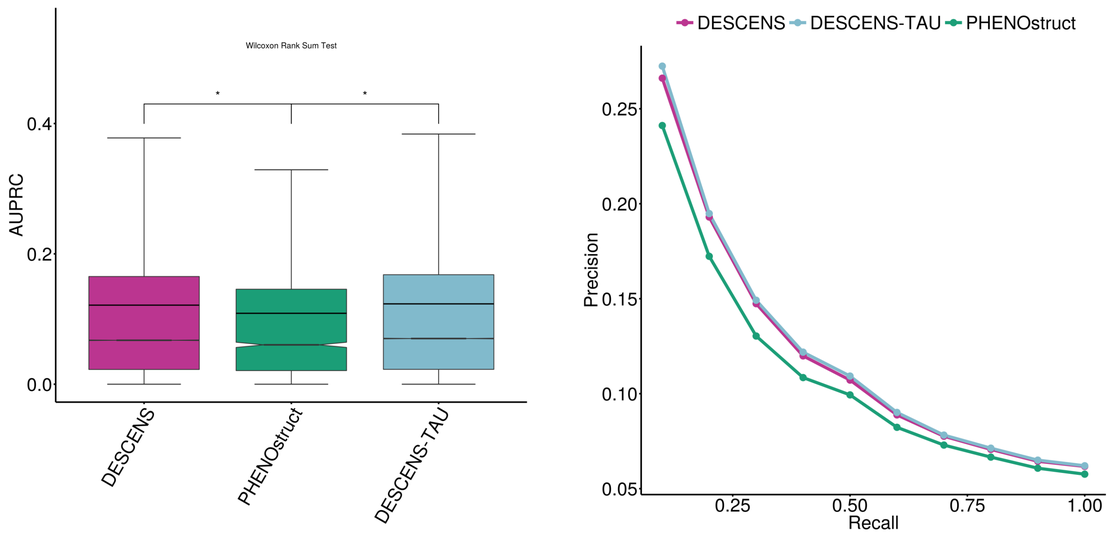
[1] P. N. Robinson, et al. The Human Phenotype Ontology: a tool for annotating and analyzing human hereditary disease. The American Journal of Human Genetics. 2008;83:610–5
[2] M. Notaro, M. Schubach, P. N. Robinson, and G. Valentini. Prediction of Human Phenotype Ontology terms by means of hierarchical ensemble methods. BMC Bioinformatics. 2017;18(1), 44
[3] I. Kahanda, et al. PHENOstruct: Prediction of human phenotype ontology terms using heterogeneous data sources. F1000Research. 2015;4

PERFORMANCE TESTING

List of novel gene–phenotype associations predicted by our algorithms using HPO 04/2016 and STRING 9.1. Confirmed by literature or newer HPO (03/2017).

Gene	HPO Term	AUROC	Evidence
XRCC2	Clubbing of Toes	1.000	HPO (March 2017)
COX10	Abnormal Mitochondria in Muscle Tissue	0.9967	10.1001/jamaneurol.2013.3242
CFB	Systemic Lupus Erythematosus	0.9967	10.1016/j.imbio.2015.08.001
CAD	Abnormality of Pyrimidine Metabolism	0.9951	10.1093/hmg/ddv057
LIPE	Insulin-Resistance Diabetes Mellitus	0.9934	HPO (March 2017)
TGFBR3	Emphysema	0.9785	10.1165/rcmb.2008-0427OC
IGF2	Neoplasm of the Adrenal Gland	0.9781	HPO (March 2017)
ECHS1	Abnormality of Fatty-Acid Metabolism	0.9753	10.1002/humu.22730
BARD1	Nephroblastoma aka Wilms Tumor	0.9615	10.1016/j.ebiom.2017.01.038

Comparison on performance of our algorithms to PHENOstruct [3]. Algorithms were trained on HPO 04/2016 with STRING 9.1 and tested to HPO 03/2017.



SOFTWARE TUTORIAL

All algorithms are available through the open-source software library **HEMDAG** and are implemented in R. Our software can be easily installed using bioconda:

```
> conda install -c bioconda r-hemdag
```

All **HEMDAG** methods are fully documented and we provide a comprehensive tutorial of the package:

https://hemdag-tutorials.readthedocs.io

Example to use DESCENS with (T) and without (TF) thresholds:

```
# Loading library and example data
> library(HEMDAG);
> data(scores,labels);
# Normalize scores to make different methods comparable
> S <- normalize.max(S);
# Select root and exclude it from computations
> root <- root.node(g);
> if(root %in% colnames(L)) L<-L[,-which(colnames(L)==root)];
> if(root %in% colnames(S)) S<-S[,-which(colnames(S)==root)];
# Run DESENCS-TF and DESENCS-T
> S.descensTF <- descens.threshold.free(S,g,root);
> S.descensT <- descens.threshold(S,g,root,t=0.5);
# Show the AUPRC using 5-fold CV
> AUPRC.single.over.classes(L, S.descensTF, folds=5,
seed=42)$average
[1] 0.6485772
> AUPRC.single.over.classes(L, S.descensT, folds=5,
seed=42)$average
[1] 0.7015372
```

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